

Teaching Statement

Teaching Philosophy

My teaching interests lie at the intersection of decision analysis, optimization, data analytics, and artificial intelligence. I am most excited to teach courses in multi-criteria decision making, engineering optimization, engineering economics and risk analysis, applied statistics, and data analytics, and to build new courses that connect these quantitative methods with AI-driven problem solving.

I believe students learn best when they can see where a course will take them. Too often, students meet a new subject as a list of formulas and only discover its purpose at the end of the semester, if at all. I begin the other way around. In the first week, I ask students what they care about: a sport they follow, a stock they hold, a product they want to build, a career they hope to have. I then use those interests to paint a concrete picture of the course: once you master these tools, here is what you will be able to do, and here is how it moves you toward your own goals.

One class session shows what this looks like in practice. Near the end of a fall semester, I covered a lecture on multi-criteria decision making (MCDM) for my advisor in Manufacturing and Automated Systems. It was the week before Black Friday, and when students saw that their regular professor was away, the room relaxed. Instead of opening my slides, I asked them about their holiday plans. Some were traveling; others were planning to shop.

One student could not decide between two monitors: one with 4K resolution, and one with a 240 Hz refresh rate but lower resolution. I asked the class to list everything that mattered to them in that choice: price, resolution, refresh rate, and anything else they cared about. Then I asked how they would decide which monitor fit them best. They quickly proposed giving each factor a weight, and just as quickly noticed that dollars, pixels, and hertz cannot be compared directly. Through discussion, they arrived at normalization on their own, and from there at a simple weighted ranking of the options from best to worst. Without seeing a single slide, they had rebuilt the most basic MCDM method.

Then something more interesting happened. Some students found that the "rational" winner was not the option they actually liked best. This opened the door to ideas beyond the syllabus. I explained that people often prefer a balanced option, or care about one outstanding feature as long as nothing else is too poor, and introduced nonlinear models,

including the Geometric Dispersion Theory approach from my own research. When a student applied the method to choosing clothes, we discussed how a subjective attribute like style can matter little when it is only slightly better, but a great deal once it is truly excellent; this led naturally to value functions. Students also asked how to assign weights more carefully and how to rank options against a reference point, so I briefly introduced AHP and reference-based methods.

The session was only meant to cover the basics, but students kept asking for more models, and I eventually had to pause the discussion to make sure everyone left with the fundamentals firmly in hand. The effect lasted beyond that day. I later supervised the course's student projects, and many teams chose MCDM topics on their own initiative; several went further and explored, without being asked, how MCDM differs from regression in what each can tell a decision maker. One team compared simple additive weighting (SAW) with ridge and lasso regression and concluded that, although the regression models predicted more accurately, SAW made them trust their final choice more, because in SAW the weights directly reflected what they considered important, while the regression coefficients were much harder to interpret in those terms. That class confirmed what I believe about teaching: when a lesson starts from something students already care about, the theory stops being something they must memorize and becomes something they want to use.

The same approach carries across every course I am prepared to teach. In Engineering Optimization, the students in that class who were planning trips could have started from their own travel budget and itinerary: what to maximize, what limits they face, and which choices they control are exactly the objective, constraints, and decision variables of a linear program. In Convex Optimization, fitting a curve to data students collected about something they care about motivates why some problems have one guaranteed best answer and others do not. In Engineering Economics, a decision students actually face, such as whether to buy a car or lease one, makes present value and risk concrete. In each case, the topic is the student's own problem first and the formal method second.

Built on this principle, I design each course in two stages. The first stage gives students solid foundations; the second teaches them to use AI to turn those foundations into projects of their own.

Stage One: Foundations as Building Blocks

I am openly enthusiastic about AI in the classroom, and I encourage students to use it. But AI can only amplify what a student already understands. A student who does not know what a constraint is cannot judge whether an AI-generated optimization model is correct; a student who has never derived a utility function cannot tell when an AI's recommendation is nonsense. Foundations are the building blocks. Without them, students have nothing to assemble; with them, they can build almost anything.

For this reason, the first half of each course focuses on theory, core formulas, and hand-worked problems. In Engineering Optimization, students formulate linear programs on paper and solve small cases by hand before touching a solver. In Engineering Economics, they compute present values and expected utilities step by step before building spreadsheets. In decision analysis, they work through a TOPSIS or AHP example manually so that every number in the final ranking has a meaning.

I keep this stage connected to the picture painted on day one. Each new concept is introduced as a piece students will need for the project they are already imagining. Short in-class exercises, frequent low-stakes quizzes, and office hours let me find gaps early, while there is still time to close them. By midterm, my goal is that every student owns a toolbox they understand well enough to check, question, and extend.

Stage Two: AI as an Amplifier, Not a Replacement

I firmly believe that learning to use AI correctly is now one of the most important skills a student can gain, and that it should be taught deliberately rather than left to chance. In the second half of my courses, I will guide students to combine their foundations with AI tools to build a project they choose. The rule I give them is simple: AI should not replace your learning; it should multiply what you can do with it. Students use AI to write code, find and clean data, build interfaces, and explore alternatives, while they remain responsible for the model, the assumptions, and the judgment behind every result.

I plan to build three practices into every project:

- **Plan before prompting.** Students first write their problem, decision variables, and criteria in their own words, using the theory from Stage One.
- **Verify, do not trust.** Every AI-generated model or result must be checked against a hand calculation, a small test case, or a sanity check students design themselves.

- **Document the collaboration.** Students will submit a short log of how they used AI, what it got wrong, and how they fixed it. This turns AI use from something hidden into a skill they can reflect on and improve.

The projects grow from the interests students named on the first day. A student who loves basketball might build a player-evaluation tool using multi-criteria decision methods; a student interested in finance might optimize a small portfolio and visualize its risk; another might design a dashboard that helps a local business compare suppliers. Each project ends with a public presentation and a polished deliverable, such as a working app, a GitHub repository, or an interactive dashboard, that students can put on their résumés and discuss in job interviews.

I expect this design to change how students relate to the course: the foundations stop feeling like hurdles and become the reason their project works. I have already seen a version of this as a Teaching Assistant. When students in Manufacturing and Automated Systems chose project topics that interested them, many went beyond the requirements on their own. Adding AI as an amplifier will let them take their ideas even further.

Teaching Experience

Since Spring 2023, I have served as a graduate Teaching Assistant for eight courses at Case Western Reserve University, spanning undergraduate and graduate levels in engineering and business:

- ECSE 346 Engineering Optimization and ECSE 416 Convex Optimization in Engineering
- ECSE 352 Engineering Economics and Decision Analysis (and its graduate counterpart, Engineering Economics and Financial Analysis)
- ECSE 360/460 Manufacturing and Automated Systems
- ECSE 350/450 Operations and Systems Design
- ECSE 408 Introduction to Linear Systems
- EPOM 405/ECSE 411 Applied Engineering Statistics
- BUAI 445 Advanced Marketing Analytics (Weatherhead School of Management)

In these roles I went beyond grading. In Manufacturing and Automated Systems, I co-designed the multi-criteria decision making module, delivered lectures on it, and supervised the student projects. In Engineering Economics, I designed the risk decision module and taught its lectures. In Operations and Systems Design, I reviewed student

stock-forecasting projects built with neural networks, ARMA, and support vector regression, and saw firsthand how much more students invest when the project is about something they care about.

Having taken several of these courses as a student before teaching them, I remember where the material felt abstract and where it finally clicked. That memory shapes how I explain concepts today.

Courses I Am Prepared to Teach

I am ready to teach existing undergraduate and graduate courses in engineering optimization, convex optimization, engineering economics, decision analysis, operations and systems design, applied statistics, and linear systems. I would also like to develop two new courses:

- **Multi-Criteria Decision Making with AI.** A course that teaches classical methods (AHP, TOPSIS, VIKOR, utility theory, and Geometric Dispersion Theory from my own research) and then has students build AI-assisted decision tools for real problems, from site selection to product ratings.
- **AI-Augmented Engineering Projects.** A project-based course for upper-level students in which they apply foundations from earlier coursework and use AI tools to design, build, and present a working system, with explicit training in verifying and documenting AI output.

Assessment, Inclusion, and Mentoring

My assessment follows the two stages. In Stage One, short quizzes and closed-book exams check that students truly own the foundations without AI support. In Stage Two, projects are graded on the quality of the problem formulation, the soundness of verification, the reflection log, and the final presentation, not on how polished the AI-generated code looks.

Starting from student interests is also how I make the classroom inclusive. When students choose project topics from their own lives, students from very different backgrounds see themselves in the course. I hold regular office hours, give early feedback before grades are at stake, and help students scope projects that stretch them without overwhelming them. For graduate students, I hope to mentor them the way my own advisor mentored me: by connecting their research to problems they care about and helping them build a portfolio of work that opens doors.

Closing

My goal as a teacher is to help students leave each course with two things: a solid understanding of the foundations, and something they built with those foundations that they are proud to show others. AI makes the second part more achievable than ever, but only if the first part is strong. I look forward to helping students learn to use these tools wisely, and to watching them turn their own ideas into real work.